

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

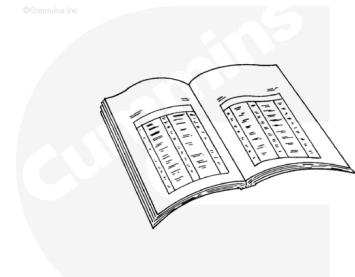
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

⚠ WARNING ⚠

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ck800wa

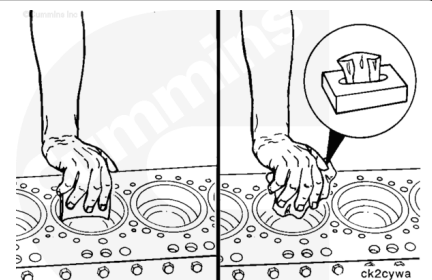
Do not rotate the crankshaft with loose or missing connecting rod caps. Damage to the piston cooling nozzles may occur.

- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/269/269-007-037-tr.html>)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/269/269-008-018-tr.html>)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/269/269-007-025-tr.html>)
- Remove the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/269/269-007-035-tr.html>)
- Remove the block stiffener plate. Refer to Procedure 001-089 in Section 1. (</qs3/pubsys2/xml/en/procedures/269/269-001-089-tr.html>)
- Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (</qs3/pubsys2/xml/en/procedures/269/269-001-046-tr.html>)
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/269/269-002-004-tr.html>)

Remove

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, following the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

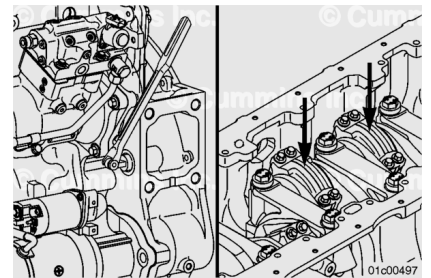
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ CAUTION ⚠

Do not use an emery cloth or sandpaper to remove carbon from the cylinder liners. Aluminum oxide or silicon particles from emery cloth or sandpaper can cause serious engine damage. Do not use any abrasives in the ring travel area. The cylinder liner can be damaged.

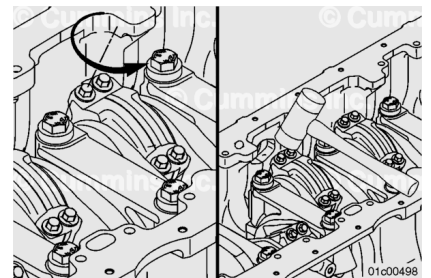
Use a fine, fibrous, abrasive pad such as Scotch-Brite™ 7448, Part Number 3823258 or equivalent, and solvent to remove carbon from the cylinder liners as needed.

Rotate the crankshaft so that two of the connecting rods are at bottom dead center.



⚠ CAUTION ⚠

Do not damage the fractured split surface on the connecting rod or connecting rod cap while the connecting rod caps are removed. If the fractured split surface is damaged, the connecting rod and connecting rod cap must be replaced to reduce the possibility of engine damage. Incorrect assembly can damage the rod.



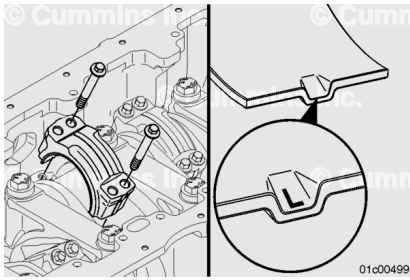
Do **not** remove the capscrews at this time.

Loosen the connecting rod capscrews.

Hit the connecting rod capscrews with a rubber hammer to loosen the rod caps from the dowels.

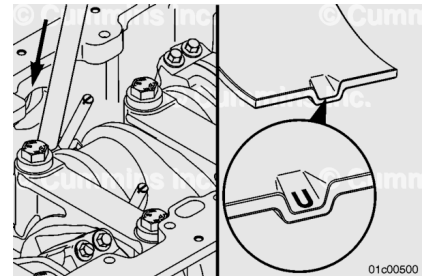
Remove the capscrews and the rod cap.

Remove the bearing shell from the rod cap and mark the cylinder number and the letter "L" on the flat surface of the bearing tangs.



Use a T-handle piston pusher to push the rod up far enough to allow the upper bearing shell to be removed.

Remove the bearing shell and mark the cylinder number and the letter "U" on the flat surface of the bearing tangs.

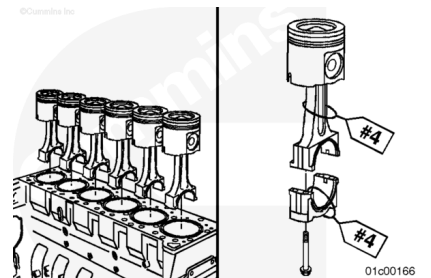


Use both hands to remove the piston and connecting rod assembly.

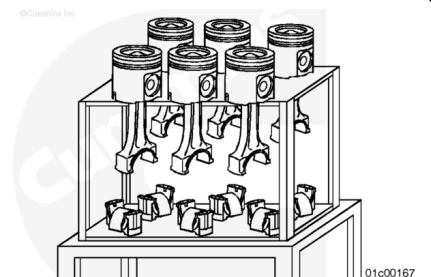
The piston and connecting rod assemblies **must** be installed in the same cylinder they were removed from to provide for proper fit of worn mating surfaces if parts are reused.

Use a tag to mark the cylinder number from which each piston and connecting rod assembly was removed.

Pistons that are reused **must** be installed in the same orientation as when removed from the engine. Mark the front side of the piston and piston pin with layout ink, or equivalent, after removal.



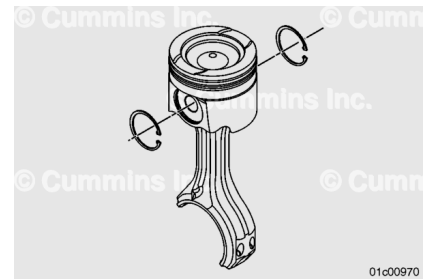
Place the connecting rod and piston assemblies in a container to protect them from damage.



Disassemble

Use internal snap ring pliers to remove the snap rings from both sides of the piston.

Note : If the piston is being replaced the piston pin **must** also be replaced.

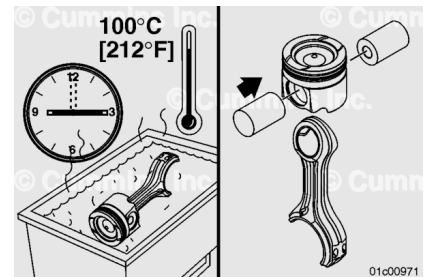


⚠ CAUTION ⚠

Do not use a hammer to remove the piston pins. The piston can distort, causing it to seize in the liner.

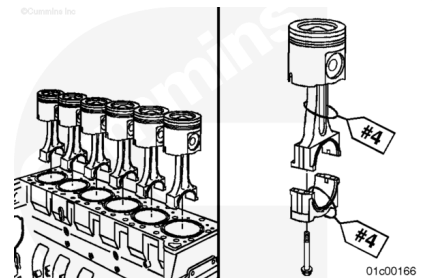
If the piston pin can **not** be easily removed by hand, place the piston and connecting rod assembly in a container of water. Heat the piston in boiling water for 15 minutes.

Use a blunt tool to push the piston pin from the piston and connecting rod assembly.



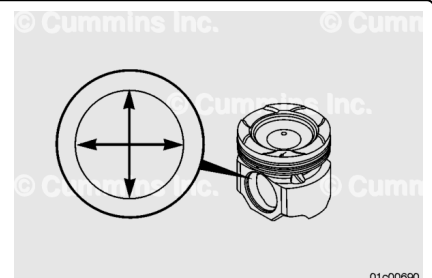
The piston and connecting rod assemblies **must** be installed in the same cylinder from which they were removed to provide for proper fit of worn mating surfaces if parts are reused.

Use a tag to mark the cylinder number from which each piston and connecting rod assembly was removed.

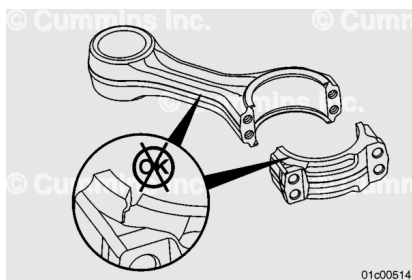


Clean and Inspect for Reuse

Use the following procedure to clean and inspect the pistons and piston pins. Refer to Procedure 001-043 in Section 1. (</qs3/pubsys2/xml/en/procedures/269/269-001-043.html>)



Use the following procedure to clean and inspect the connecting rods. Refer to Procedure 001-014 in Section 1. (</qs3/pubsys2/xml/en/procedures/269/269-001-014.html>)



Assemble

⚠ CAUTION ⚠

The retainer snap ring must be seated completely in the piston pin groove to prevent engine damage during engine operation.



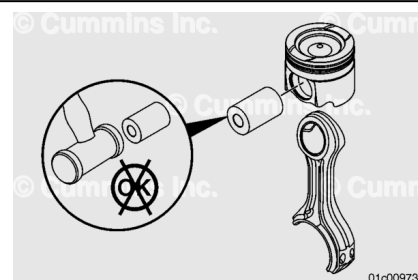
Install a new snap ring into one piston pin bore of each piston.

Install the snap ring with the gap at the top or bottom of piston pin bores.

If the pistons are being reused, the piston and pin **must** be matched to the way they were when they were removed.

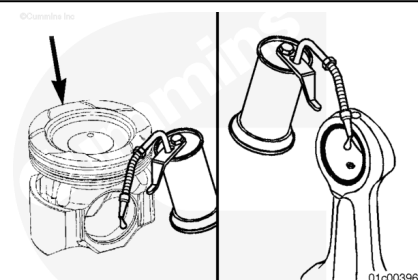
⚠ CAUTION ⚠

Do not use a hammer to install the piston pin. The piston can distort, causing it to seize in the liner.



⚠ CAUTION ⚠

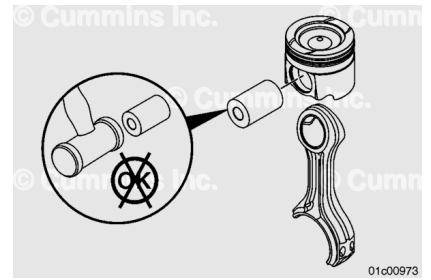
Generously coat both piston pin bores with a heavy weight gear oil, such as 85W-140 EP, when assembling pistons. Also lubricate the connecting rod bushing. If gear oil is not available, Lubriplate™, Part Number 3163086, is permissible. Failure to adequately lubricate the pin joint can result in extensive engine damage.



Use clean 85W-140 oil to coat the piston pin.

⚠ CAUTION ⚠

The piston must be installed onto the connecting rod with the correct orientation or engine damage can occur.



For pistons that are being reused, the piston, piston pin, and connecting rod **must** be installed in the same orientation as when were removed from the engine.

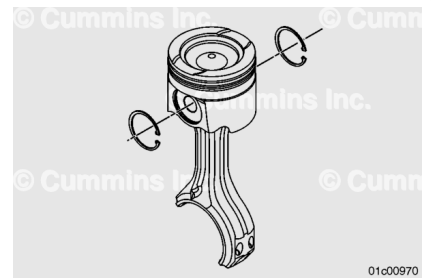
New pistons are symmetrical and do not require a specific orientation during installation.

Note : Pistons do not require heating to install the pin, however, the pistons do need to be at room temperature or above.

Install the pin through the piston and connecting rod.

⚠ CAUTION ⚠

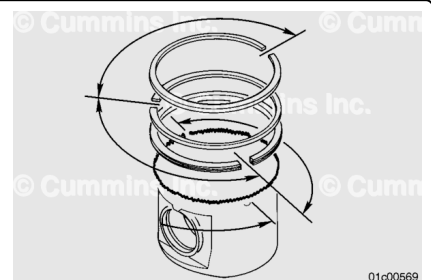
The snap ring must be seated completely in the piston groove to prevent engine damage during engine operation.



Install a new snap ring into the piston pin bore.

Install the snap ring with the gap at the top or bottom of piston pin bore.

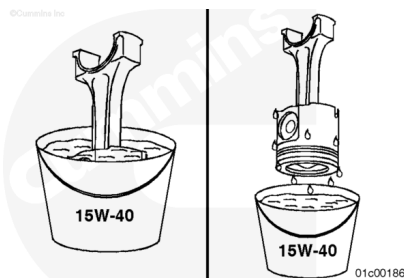
Install the piston rings. Refer to Procedure 001-047 in Section 1. (</qs3/pubsys2/xml/en/procedures/269/269-001-047-tr.html>)



Install

Immerse the piston in a container of clean 15W-40 oil.

Remove the piston from the container. Let the excess oil drain from the piston.



Use a clean, lint-free cloth to clean the connecting rods and bearing shells.

Do **not** lubricate the backside of the bearing shells. The operating clearance of the bearing will be reduced and the bearing can be damaged during engine operation.



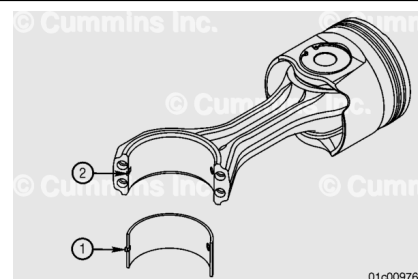
If new bearings are **not** used, the used bearings **must** be installed on the same connecting rod from which they were removed.

Install the upper bearing shell into the connecting rod.

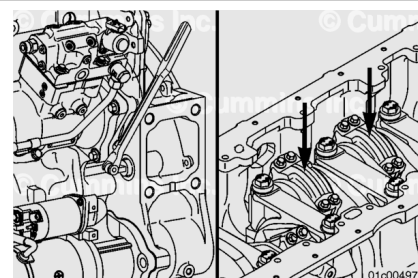
The tang of the bearing shell (2) **must** be in the slot of the cap (1). The end of the bearing shell **must** be even with the cap mounting surface.

Use Lubriplate™ 105, Part Number 3163086 or equivalent, to coat the entire inside circumference of the bearing shell.

Apply a film of clean 15W-40 oil to the cylinder liner.

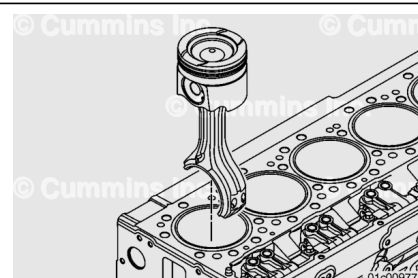


Use the barring adapter on the flywheel housing or $\frac{3}{4}$ " drive on the damper to rotate the crankshaft so the connecting rod journal of the connecting rod being installed is at bottom dead center. Refer to Procedure 000-017 in Section 0.
(/qs3/pubsys2/xml/en/procedures/269/269-000-017.html)



Install the piston and ring assembly in a container of clean 15W-40 oil.

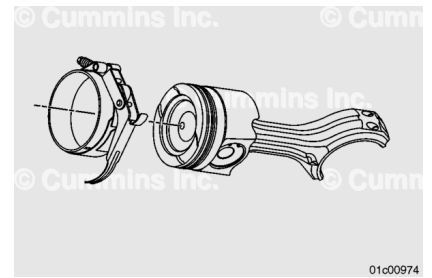
Remove the piston and ring assembly from the container, and let the excess oil drain from the piston.



Inspect the cylinder liners for reuse. Refer to Procedure 001-028 in Section 1. (/qs3/pubsys2/xml/en/procedures/269/269-001-028-tr.html)

Use piston ring compressor, Part Number 4919273, to compress the rings.

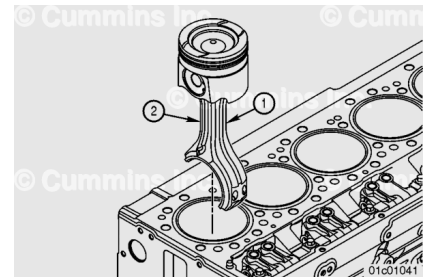
The cylinder block and all parts **must** be clean before assembly.



For pistons that are being reused, the piston, piston pin, and connecting rod **must** be installed in the same orientation from which they were removed from the engine.

New pistons are symmetrical and do not require a specific orientation during installation.

The long side of the connecting rod will be on the left side of the engine and the short side of the connecting rod will be on the right side of the engine when properly installed.

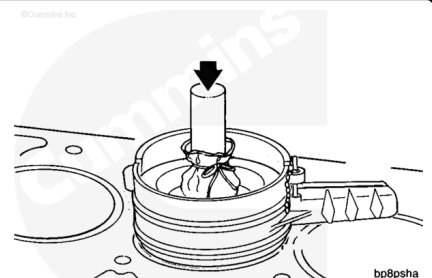


⚠ CAUTION ⚠

Do not use a metal drift to push the piston into the cylinder liner. The piston rings or cylinder liner can be damaged.

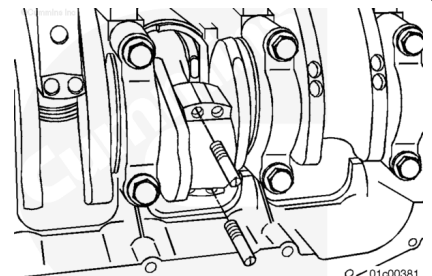
Hold the ring compressor against the cylinder liner. Push the piston through the ring compressor and into the cylinder liner. Push the piston until the top ring is completely in the cylinder liner.

If the piston does **not** move freely, remove the piston and inspect for broken or damaged rings.



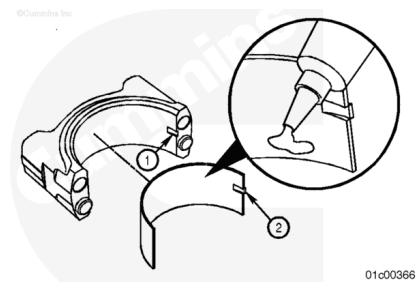
Align the connecting rod with the crankshaft while pushing the piston and connecting rod assembly into place.

Remove the nylon guide pins.



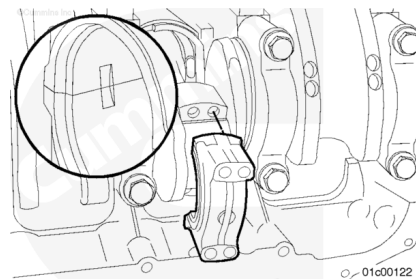
Install the bearing shell into the connecting rod cap with the tang (2) of the bearing in the slot (1) of the cap.

Use clean Lubriplate™ 105 multi-purpose lubricant, or its equivalent, to lubricate the bearing shell to crankshaft journal mating surface.



⚠ CAUTION ⚠

Do not damage the fractured split surface on the connecting rod or connecting rod cap while the connecting rod caps are removed. If the fractured split surface is damaged, the connecting rod and connecting rod cap must be replaced to reduce the possibility of engine damage. Incorrect assembly can damage the rod.

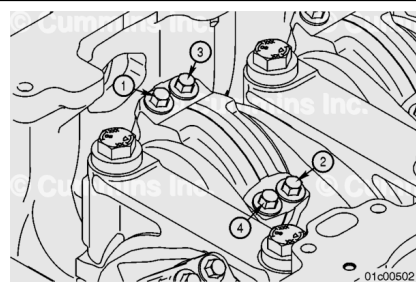


Install the cap onto the connecting rod.

Lubricate the threads and underside of the head of the connecting rod capscrews with clean engine oil.

Thread the capscrews into the connecting rod and hand tighten.

Tighten the capscrews in numerical sequence.

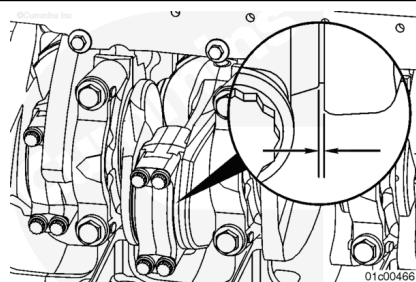


Torque Value:

1. 75 n•m [55 ft-lb]
2. Loosen all capscrews
3. 75 n•m [55 ft-lb]
4. Rotate the capscrews in numerical sequence 60 degrees.

Measure the connecting rod side clearance.

Connecting Rod Side Clearance		
mm		in
0.07	MIN	0.0027
0.30	MAX	0.0118



The connecting rod **must** move freely from side to side on the crankshaft journal. If the rod does **not** move freely, remove the rod cap and make sure the bearing shells are the correct size.

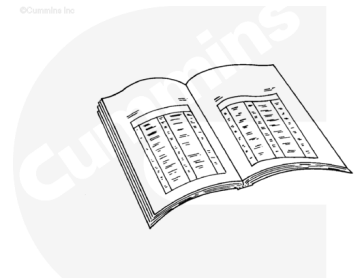
Check for dirt or damage on the crankshaft and the bearing shells.

Repeat the above steps to install the remaining bearing shells and connecting rod caps.

Finishing Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



ck800wa

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

- Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1.
(/qs3/pubsys2/xml/en/procedures/269/269-001-046-tr.html)
- Install the block stiffener plate. Refer to Procedure 001-089 in Section 1.
(/qs3/pubsys2/xml/en/procedures/269/269-001-089-tr.html)
- Install the lubricating oil suction tube. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/269/269-007-035-tr.html)
- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/269/269-007-025-tr.html)
- Install the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/269/269-002-004-tr.html)
- Prime the lubricating oil system. Refer to Procedure 007-037 in Section 7.

(/qs3/pubsys2/xml/en/procedures/269/269-007-037-tr.html)

- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/269/269-008-018-tr.html)
- Fill the lubricating oil system. Refer to Procedure 007-037 in Section 7. (/qs3/pubsys2/xml/en/procedures/269/269-007-037-tr.html)
- Operate the engine to normal operating temperature and check for leaks.